

Hector Lugo III

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EDUCATION

The University of Texas Rio Grande Valley

Master of Science in Computer Science

Aug. 2024 – Dec. 2026

GPA: 3.79/4.00

The University of Texas Rio Grande Valley

Bachelor of Science in Computer Science

Aug. 2020 – May 2024

EXPERIENCE

Summer Research Intern

Air Force Research Laboratory (AFRL), Rome, NY

May 2026 – Aug. 2026

- Selected for the 2026 Department of War HBCU/MI Summer Research Internship Program at AFRL, conducting full-time research alongside DoW STEM professionals.
- Developing reinforcement learning policies and vision-language models for autonomous systems; presenting research outcomes at the DoW Intern Symposium in Arlington, VA.

Machine Learning Engineer Intern

Idaho National Laboratory (INL)

May 2025 – Present

- Engineered a ResNet Autoencoder to denoise TDMS signals, significantly improving data quality for smart grid anomaly detection.
- Accelerated model training time by over 93% (from 30 days to 1–2 days) utilizing Fast Fourier Transform (FFT) preprocessing and HPC (Slurm).
- Authored comprehensive technical documentation for the denoising pipeline, ensuring reproducibility and long-term clarity for all project contributors.

Research Assistant

Machine Intelligence Lab, UTRGV

Jan. 2023 – Present

- Conducting Vision-Language Model (VLM) and Vision-Language-Action (VLA) research for humanoid locomotion and control using LocoMujoco.
- Engineered custom simulation environments within Isaac Sim to train scalable reinforcement learning policies for Unitree Go2 autonomous navigation.
- Designed autoencoders for data compression, reducing input dimensionality and compute load while accelerating convergence in continuous control tasks.

Graduate Teaching Assistant

UTRGV

Jan. 2024 – Present

- Instruct CS 1101 (Intro to CS) and Digital Image Processing, mentoring 40+ students across lectures, labs, and grading assignments.
- Guided 100+ students in assembly design, testing, and code reviews, strengthening low-level programming fundamentals.

Summer Research Intern

University of California Riverside

June 2023 – Aug. 2023

- Applied spectral clustering to U.S. railway accident data, identifying critical spatial safety zones and improving the accuracy of predictive safety and risk assessment models.
- Published research findings on spatial safety analysis at the ASME JRC2024 Conference.

PROJECTS

Personalized Chemotherapy Dosing | *Decision Transformer, Offline RL*

Jun. 2024 – Jul. 2024

- Developed a Decision Transformer model for personalized dose control utilizing state-conditioned inputs.
- Accepted at ICICT 2025 for contributions to adaptive treatment planning via offline reinforcement learning.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JavaScript, HTML/CSS, R

Frameworks & Tools: PyTorch, NumPy, Pandas, FastAPI, React, Flask, Git, Docker, Linux, HPC (Slurm)

Robotics & Simulation: Omniverse Isaac Lab, Isaac Sim, MuJoCo, LocoMujoco, ROS, Unitree Go2 Ecosystem

Concepts: Reinforcement Learning, VLM, VLA, Computer Vision, Signal Processing, Autoencoders, FFT, Decision Transformers